

ABNORMALITY REPORT — RCA & CAPA/PAA

Equipment Related Risk: Cooling Water Pump PM-4405B (Motor Driven)

Problem Statement

Cooling Water Pump PM-4405B motor tripped on 08-Jul-2026 due to drive-end bearing failure from grease degradation and an over-extended re-lubrication interval.

General Identification & Details

Tag / Plant: PM-4405B | NUP (Nova Utility Plant)

Pre-Risk Level: III (Score 200)

AR Type: OPEDR (Operation) — Uptime

Severity: Moderate

Discipline: ELE (Electrical)

Downtime: 8.0 hours

Date Occurrence: 08 Jul 2026

Financial Loss: \$112.0k

Date Reported: 08 Jul 2026

PIC (RCA): REL-02

Immediate Action & Procedure

Immediate Action Taken: Start standby CW pump PM-4405C, isolate PM-4405B, replace motor DE & NDE bearings.

Procedure Reference: Procedure No. RCA-P-0050-03 • Form RCA-F-0075-02

DOWNTIME IMPACT

8.0 h

ESTIMATED LOSS

\$112.0k

Cooling Water Pump PM-4405B (Motor Driven) — Motor Bearing Failure (Overheating)

Tag / Eq. Class: PM-4405B | Class B

Chronology of Events

Time / Date	Event Description
06-Jul-2026	Cooling-water system running on PM-4405B; motor DE bearing temperature trending up (68 → 84 °C).
08-Jul-2026 13:20	Motor bearing high-temperature alarm; motor tripped on winding/bearing over-temp.
08-Jul-2026 13:35	Standby PM-4405C started; cooling-water supply maintained at reduced margin.
08-Jul-2026 16:00	Motor DE bearing found seized; grease degraded/carbonized. NDE bearing marginal.
08-Jul-2026 21:20	DE & NDE bearings replaced and re-greased; motor recommissioned.

Past Performance Analysis

Historical Data & Evidence: PM-4405B motor bearing temperature baseline is 60–68 °C. Re-lubrication was performed on a fixed 12-month interval regardless of duty/ambient. Bearing temperature was not trended in CMMS, so degradation went unnoticed.

Impact Summary (4W):

- **What:** PM-4405B motor DE bearing failed / overheated
- **When:** 08-Jul-2026 (13:20)
- **Scope:** Loss of one cooling-water pump; reduced CW margin (standby covered)
- **Uptime Loss:** 8.0 hours → 160.0 ton → \$112.0k

Target Setting

ACTUAL CONDITION (Before Improvement) CW pump motor trip due to bearing overheating.	TARGET CONDITION (Project Y) No motor bearing failure due to lubrication (bearing temp < 75 °C, risk-based re-greasing).
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Recurrence of this equipment failure mode:
Before = 1 case → Target = 0 case

Root Cause Analysis: Parameter Verification (4P)

Parameter	Result	Evidence / Finding
P1: Motor bearing temperature above limit	NG	DE bearing reached 84 °C vs 75 °C alarm — grease breakdown.
P2: Motor over-loaded (high ampere)	G	Running current 132 A vs 150 A rated — within limit.
P3: Cooling-fan / ventilation blocked	G	Fan cowl clean; ambient ventilation adequate.
P4: Grease type incorrect	G	Correct high-temp lithium-complex grease used.
P5: Bearing beyond service life	G	Bearing in service 2 years vs 5-year design — within life.

NG = current condition not meeting standard (problem) • G = meets standard

Root Cause Analysis: 4M + 1E Verification

Factor	Result	Evidence / Finding
X1: Re-lubrication interval not risk-based	NG	Fixed 12-month interval too long for duty & ambient — grease degraded.
X2: Bearing temperature not trended in CMMS	NG	No condition-based trigger; only alarm at failure point.
X3: Vibration route excludes utility motors	NG	PM-4405B not on the monthly vibration route.
X4: Operator round frequency	G	Rounds performed per schedule; local gauge read.

ROOT CAUSE

Motor drive-end bearing failure caused by grease degradation from an over-extended, non-risk-based re-lubrication interval, undetected due to absence of bearing-temperature trending, resulting in bearing overheating and motor trip.

Corrective & Pro-Active Action (CAPAA)

Corrective / Pro-Active Action	Plan Date	PIC	Status
X1: Revise PM-4405B re-lubrication interval to risk-based (4-monthly)	31-Jul-2026	REL-02	In Progress
X2: Add motor bearing temperature trending & alert to CMMS	31-Aug-2026	REL-05	Open
Pro-Active: X1: Apply risk-based re-greasing to all CW pump motors	31-Aug-2026	REL-02	Open
Pro-Active: X2: Extend bearing-temp trending to utility motor fleet	30-Sep-2026	REL-05	Open

Preventive & Risk Analysis

Item / Root Cause	Preventive Action / Countermeasure	Plan Date	PIC
X3: Motor not on vibration route	Add PM-4405B/C to monthly vibration & thermography route	31-Jul-2026	REL-02
P1: Bearing over-temp	Verify motor cooling-fan & louvre condition each PM	31-Jul-2026	ELE-01
Shorten re-greasing interval	Risk: Over-greasing risk (churning / seal push-out) Countermeasure: Use metered re-greasing per bearing volume calc	31-Jul-2026	REL-02

PM Schedule Established

- PM-1: Risk-based motor bearing re-greasing PM-4405B (Electrical, 4-monthly)
- PM-2: Monthly motor bearing temperature & vibration reading (Reliability, Monthly)

Impact vs Control Prioritization

CONTROL / IMPACT	Low	Medium	High
Low			
Medium			X1 X2 X3
High			P1

Downtime & Closure Summary

DOWNTIME	PRODUCTION LOSS	ESTIMATED LOSS	PRE-RISK
8.0 h	160.0 ton	\$112.0k	III (200)

KEY CORRECTIVE / PRO-ACTIVE ACTIONS

- Revise PM-4405B re-lubrication interval to risk-based (4-monthly)
- Add motor bearing temperature trending & alert to CMMS
- Add PM-4405B/C to monthly vibration & thermography route